FermentAI Sourdough

The recipe behind the predicted slices: two identical mini-loaves, two fermentation chambers

Mix one dough twice. Ferment one at **22 °C** (Chamber A) and the other at **27 °C** (Chamber B), then bake, slice and taste them blind. FermentAI's model predicts how each will rise and sour, and what its crumb might look like. This sheet tells you what to do, what the model expects and when, and where to write down what really happened.

Ingredients: make this twice (one dough per chamber)

Ingredient (per loaf)	Weight
Bread flour	200 g
Water	135 g
Ripe starter (100 % hydration)	40 g
Salt	4.4 g
Total dough	**~379 g**

Baker's math	
Total flour (incl. starter)	220 g
Total water (incl. starter)	155 g
Hydration	70.5 %
Salt, % of flour	2.0 %
Starter, % of dough	10.5 %

Both doughs must be identical: same flour, hydration, starter, feeding history, inoculation and mixing. The only planned difference is the chamber temperature.

The two chambers

Chamber A: 22 °C

Mild hypothesis. Cooler fermentation (22 °C start) gives relatively mild acidity with sufficient expansion.

Caveat: Long cool fermentations can still develop pronounced (often sharper, more acetic) sourness; mildness is not guaranteed.

Chamber B: 27 °C

More-acidic hypothesis. Warmer fermentation (27 °C start) gives faster acidification and a detectably more acidic loaf.

Caveat: 12-16 h at 27 °C can overproof a small dough; end fermentation or lower the temperature when the rise peaks or starts to fall, rather than holding 27 °C blindly.

Planned observation window: 12–16 h, covering fermentation and proofing as appropriate. Both setpoints are **hypotheses**, not guaranteed flavour outcomes.

Before you bake: safety and caveats

- **Young starter.** The starter is only 4-5 days old (young; may not yet be mature). Do not start until it passes the readiness checklist on page 2.
- **One experimental pH probe.** There is a single probe and permanent immersion is experimental. Take readings from a separate, non-edible sample if the probe is not food-contact certified; calibrate and clean it before trusting readings in thick dough. pH will not be continuous for both chambers.
- **Printed crumb specimens are display-only.** The 3D-printed resin slices are never food-safe: keep them away from bread, dough and starter.
- **Temperatures are hypotheses.** Warmer is expected to be more acidic, but flavour is not guaranteed, and a long cool ferment can still turn sharply sour.
- **Every timing and crumb number in this sheet is a model prediction, not a measurement.** Judge the dough by eye and touch; a human makes the bake-readiness call.

Step 0: is the starter ready?

Tick every box before mixing. Check over at least two consecutive feeds.

- ☐ Reliably doubles within approximately 6-8 h after feeding (check over at least 2 consecutive feeds).
- ☐ Contains abundant bubbles throughout, not just on the surface.
- ☐ Clean fermented aroma (yeasty / lactic / mildly acetic), no nail-polish or putrid odour.
- ☐ No visible mold or abnormal discoloration (pink, orange or fuzzy growth means discard).

If it is not ready — CONTINGENCY - disclose in all reports

Why it matters: An immature starter gives slow, unpredictable rise and acidification; the kinetic model's starter-carry-over assumptions (yeast/LAB inoculum, initial acids) would not hold.

1. Use a mature starter from another source (preferred; keeps the experiment a sourdough comparison).
2. Spike both doughs identically with a small amount of commercial yeast; must be disclosed and labelled, because it changes leavening kinetics and the yeast:LAB balance.

Method

1. **Feed the starter.** Feed the mother starter and confirm it doubles within about 6–8 h before relying on it.
2. **Mix two identical doughs.** For each dough combine 200 g flour, 135 g water, 40 g ripe starter and 4.4 g salt. Same ingredients, same order, same mixing for both. Label them A and B.
3. **Rest.** Cover and rest 20–30 min.
4. **Fold.** Develop the gluten with several sets of folds over the first 1–2 h.
5. **Into the chambers.** Put each dough in its own covered, food-safe vessel: A into the 22 °C chamber, B into the 27 °C chamber. Where helpful, put a small representative sample in view of the side camera for rise measurement. Note the clock time: this is **hour 0** on page 3 and in the bake log.
6. **Log as you go.** Record chamber temperature, rise and periodic pH (single probe, separate non-edible sample) in the bake log on page 5. Note every time a chamber door is opened.
7. **Watch the dough, not the clock.** Use the page 3 table as a guide only. End bulk fermentation, or lower the temperature, when the rise plateaus or starts to fall. Do not hold the setpoint blindly; Chamber B is the one most at risk of overproofing.
8. **Shape and proof.** Shape each loaf and proof it when it is ready (domed, airy and jiggly, springs back slowly), not by elapsed time alone.
9. **Bake.** About 450 °F / 232 °C: covered for 18–20 min, then uncovered for roughly 15–20 min. Adjust for the mini-loaf size and your oven. Keep the loaves labelled.
10. **Cool, slice, photograph, taste.** Cool fully before slicing. Slice each loaf through the middle, photograph the cut face next to the predicted slice (page 4), then hold a blind tasting of sourness, texture and aroma.

What to expect, and when to act

Model prediction, not a measurement. Times are hours after the doughs enter their chambers (hour 0), from a 128-sample simulation (provenance: *simulated*). Real dough, flour and a young starter can run faster or slower: act on what you see.

Predicted milestone	Chamber A — 22 °C	Chamber B — 27 °C
Dough has doubled (2×)	~7.2 h	~5.7 h
Peak rise	~2.11× at ~9.5 h	~2.09× at ~7.2 h
Overproof risk begins (5 % of the rise lost)	~12.9 h	**~9.5 h**
pH at 12 h / 16 h	4.31 / 4.13	4.02 / 3.90
Rise at 12 h / 16 h	2.07× / 1.97×	1.95× / 1.81×
Lactic / acetic acid at 12 h (g/kg)	3.7 / 0.69	5.5 / 0.91
Lactic / acetic acid at 16 h (g/kg)	4.7 / 0.88	6.5 / 1.07

Paired comparison over the same 128 parameter draws: B is more acidic at 12 h in 100 % of draws (median pH difference -0.25, 5–95 % range -0.41 to -0.15) and peaks earlier in 100 % (median 2.4 h earlier). The direction is robust in the model; the size of the difference is not.

Chamber A: what to do

- Start checking rise closely from ~6.2 h; expect it to double around ~7.2 h.
- Peak predicted around ~9.5 h. When the rise plateaus, end bulk and shape.
- **Shape before ~12.9 h**: after that the model predicts the dough starts losing rise (down to 1.97× by 16 h).
- Holding 22 °C needs no planned temperature change.

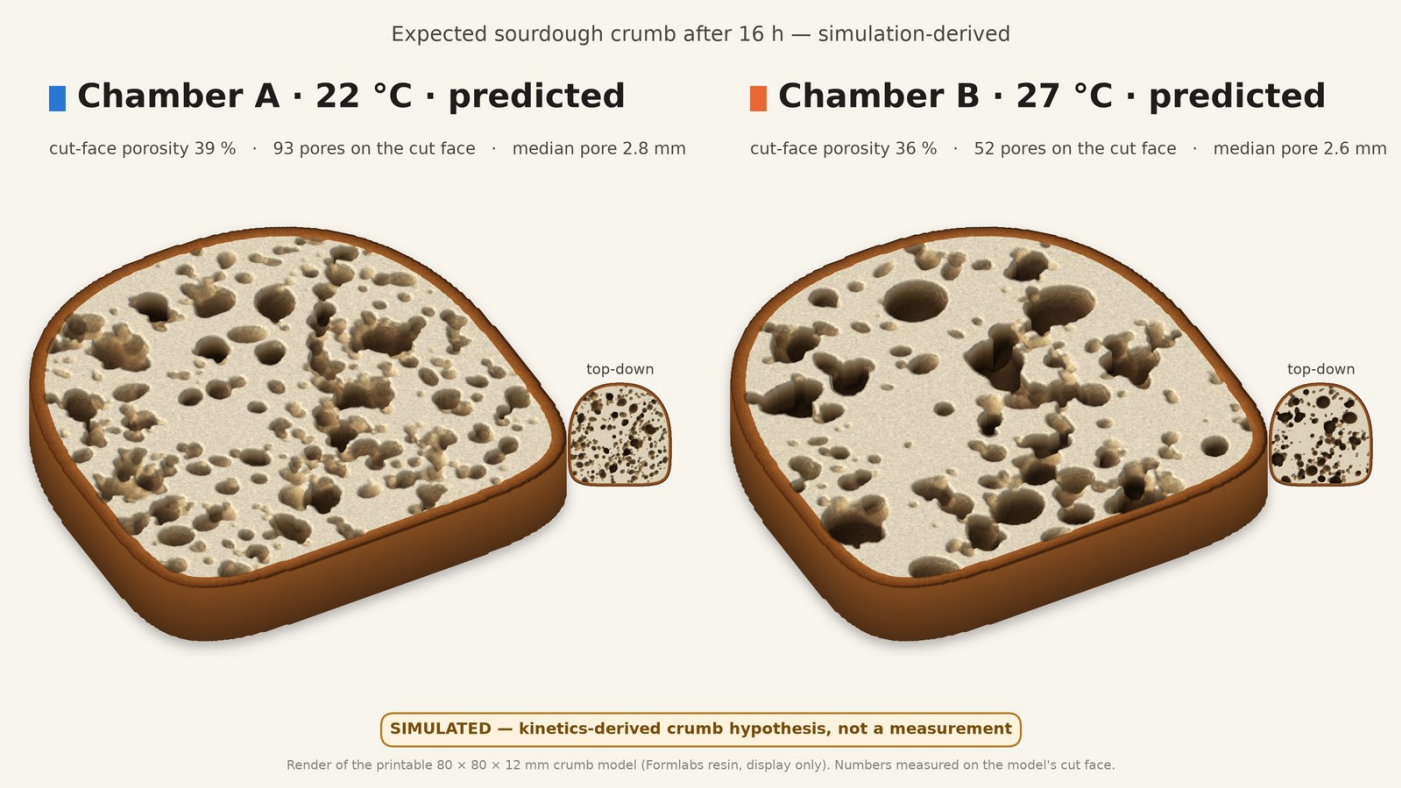
Chamber B: what to do

- Start checking from ~4.7 h; expect it to double around ~5.7 h and peak at ~7.2 h.
- **Holding 27 °C for the full 12–16 h window is predicted to overproof B** (rise lost from ~9.5 h; 1.81× by 16 h).
- **Recommended:** cool B to 22 °C at ~6 h. Model: overproof delayed to ~11 h, pH 4.01 at 16 h.
- Gentler option: cool to 18 °C at ~4 h (model: pH 4.25 at 16 h, no overproof signal within 16 h), at the cost of less acidity.
- Or end bulk and shape as soon as the rise peaks. Log any change in the bake log.

Source of the cooling interventions: docs/scientific_assumptions.md, section 5 (model examples). The chamber plan in configs/experiments.yaml holds the setpoints constant; any change is a human or agent decision made while watching the real rise. Model predictions, not measurements.

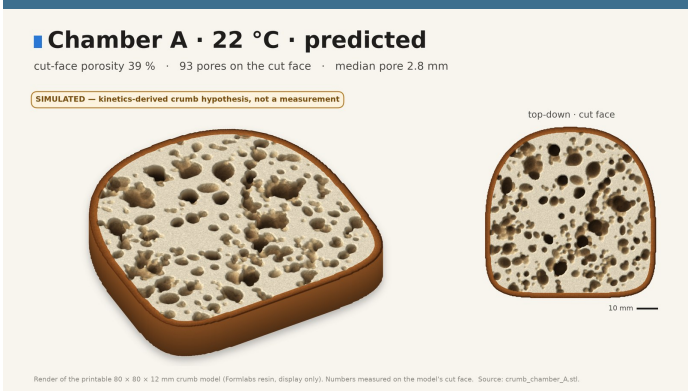
The predicted slices

Simulated hypothesis — compare with your real slice photo. These crumbs were generated from the fermentation simulation through documented, assumed mapping coefficients. Fermentation kinetics do not uniquely determine crumb structure.



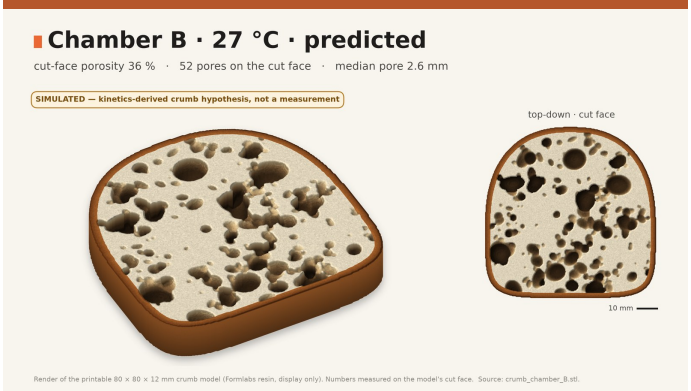
Predicted cut faces, Chamber A (left) and Chamber B (right). Rendered from the simulated meshes.

Chamber A



Cut-face porosity **39 %** · **93** pores on the cut face (1.86 per cm^2)

Chamber B



Cut-face porosity **36 %** · **52** pores on the cut face (1.04 per cm^2)

What the model expects: A, the cooler and slower dough, keeps more pores and a higher cut-face porosity; B, fermented warmer and closer to overproof, ends with fewer pores (52 vs 93) and a lower porosity. Numbers are measured on the simulated mesh (data/stl/chain_summary.json).

How to compare: slice each cooled loaf through the middle, photograph the cut face flat-on with even light and a ruler in frame, label it A or B, and set it next to the matching image above. Note anything that differs (open vs tight crumb, big tunnels, dense gummy layer).

Bake log

Hour 0 = doughs into chambers. Log at least every hour, plus at every fold, temperature change, shaping and bake.

Hour 0 clock time: _____ Date: _____ Baker: _____

Chamber A — 22 °C start							
Clock time	Hour	Chamber °C	Starter / dough °C	Rise / height	pH	Door open?	Notes

Chamber B — 27 °C start							
Clock time	Hour	Chamber °C	Starter / dough °C	Rise / height	pH	Door open?	Notes

Blind tasting — type these into the dashboard's *"What did the bread taste like?"* panel (free text, plus optional notes per loaf).

Loaf A: sourness / texture / aroma

_____

_____

Loaf B: sourness / texture / aroma

_____

_____

Which was more sour? A / B Judge: _____ What would you change next time?

_____